

Abhinav Jain

abhinav98jain@gmail.com | linkedin.com/in/abhinavj98 | +1(541)908-3982

SUMMARY

Soon-to-be Ph.D. graduate in Robotics and AI researcher specializing in complex **simulation design** using **Isaac Sim and Labs** and sim-to-real transfer for unstructured, real-world orchard environments. Expertise in applying **Deep Reinforcement Learning** and **Imitation Learning** to deploy robust manipulation policies on physical robotic platforms. Recognized for research contributions in top-tier conferences (**ICRA, CVPR, AAAI**), with proven experience in synthetic data generation and training RL, generative AI and Large Language Models (LLMs).

EDUCATION

Oregon State University

Sep 2021 - Sep 2026

PhD, Major - Robotics. Minor - Artificial Intelligence

Corvallis, OR, USA

Advisors: [Dr. Cindy Grimm](#) and [Dr. Stefan Lee](#)

Courses: Deep Learning, Learning Based Control, Natural Language Processing, Kinematics/Dynamics

GPA: 3.91/4

Oregon State University

Sep 2021 - Aug 2026

MS, Major - Robotics. Minor - Artificial Intelligence

Corvallis, OR, USA

Advisors: [Dr. Cindy Grimm](#) and [Dr. Stefan Lee](#)

GPA: 4.0/4.0

Sardar Vallabhbhai National Institute of Technology

Aug 2016 – Jul 2020

Bachelor's of Technology, Electronics and Communication Engineering

Surat, India

Seminar: Generative techniques to generate artificial datasets

Capstone: Birding - Generative Adversarial Networks to convert text description to bird images.

(Language conditioned Image generation)

EXPERIENCE

[Oregon State University](#)

Dec 2021 - Present

Graduate Research Assistant - Robotic Manipulation & Control

OR, USA

- Secured funding under the [AgAID](#) grant to collaborate with stakeholders in **AI** and **Agriculture**.
- Engineered a comprehensive **learning pipeline** for **complex robotic manipulation** utilizing **simulation design, reinforcement learning**, and **synthetic data generation** for fruit trees.
- Implemented **ROS2** to deploy **learning-based control policies** and **image segmentation** models on real-world robotic systems.

Simulation

- Architected custom manipulation simulators in **PyBullet** and the **Newton Physics Engine** for **real-world tree pruning and apple picking tasks**, incorporating **domain randomization**, custom **kinematic controllers**, and stable **deformable object models**.
- Developed a **procedural 3D asset generation** tool ([TreeSim_Lpy](#)) and a **synthetic data pipeline**, successfully passing expert indistinguishability tests to advance **computer vision** and **imitation learning** tasks.

Learning

- Formulated a novel **hybrid Reinforcement Learning** method integrating **suboptimal expert data** into an **online framework**, outperforming naive PPO and **Imitation Learning** baselines by a statistically significant 6%.
- Trained and deployed a **visual servoing policy** for a UR5 manipulator using **Deep RL (PPO)**, processing **image inputs** to achieve a $> 50\%$ success rate in **unstructured physical environments**.

- (In-progress) Training a **tactile force-based policy** for a Franka FR3 manipulator using **Deep RL (PPO)** for **Deformable Object Manipulation**, apple-picking in **unstructured physical environments**.

Perception

- Generated **synthetic data** and automated the labeling of large field-collected datasets using **SAM2** and a custom **YOLO** model for **unstructured orchard environments**, improving **target fruit detection** accuracy by 27% (achieving a 96% success rate) across **varying outdoor conditions**.

Robotics

- Deployed a **learned manipulation policy** onto a physical UR5 robotic arm using **ROS2** and **MoveIt!**, reducing experimentation time by 80% via **Behavior Trees**.
- Validated **sim-to-real transfer** by benchmarking **RL policy performance** against **RRT-Connect** using **real-world point clouds**, demonstrating a 49% success rate compared to the 27% baseline.

AgRobotics

Jun 2025 - Sep 2025

Robotics Engineer Intern

CA, USA

- Developed scalable robotic simulation pipelines utilizing **NVIDIA Isaac Sim** and **Isaac Lab** for autonomous systems.

Samsung Research Institute, Bangalore

Jan 2021 - Aug 2021

Engineer

Bangalore, India

- Developed network solutions as an engineer on the 5G NR MAC team.
- Programmed **Unit Tests** and **Block Tests** utilizing the Google Test framework, increasing the test coverage metric from 2.1 to 3.2 (exceeding the required threshold of 3.0).
- Earned the Software Professional Certification for demonstrated proficiency in **Data Structures and Algorithms**.

RBCPS LAB, Indian Institute of Science (IISc)

Jul 2020 – Jan 2021

Research Intern

Bangalore, India

- Worked in Robert Bosch Centre for Cyber-Physical Systems, IISc Bangalore, under [Dr. Chiranjeev Bhattacharyya](#).
- Adapted state-of-the-art Image Inpainting models in **PyTorch** to solve Dynamic to Static LiDAR Reconstruction.
- Evaluated reconstruction quality by executing **SLAM** on the reconstructed **LiDAR** output using **ROS**.
- Formulated the “LiDAR Quality Index” metric to rank the quality of LiDAR frames without requiring reference ground truth data.

Large Language Models - Class NLP with Deep Learning

Oregon State University

- Fine-tuned an **LLM** on GSM8K math word problems utilizing a custom **reinforcement learning** algorithm [\[Documentation and Code\]](#)
- Built an **autoregressive translation model** leveraging **attention mechanisms** and created a dynamic story generator utilizing **Transformer language models** [\[Documentation and Code\]](#)
- Programmed **Recurrent Neural Networks (RNNs)** to execute accurate **Part-Of-Speech tagging** and developed **tokenizers** for complex text processing [\[Documentation and Code\]](#)

PUBLICATIONS

Abhinav Jain, Cindy Grimm, Stefan Lee. “Visuomotor Robotic Pruning in Planar Orchards Using Hybrid Reinforcement Learning” (In review) [\[Paper\]](#)

Tieqiao Wang, **Abhinav Jain**, Sinisa Todorovic, Cindy Grimm. “A Dataset for Semantic and Instance Segmentation of Modern Fruit Orchards” [CVPR V4A Workshop \(Vision for agriculture\)](#) [\[Paper\]](#)

Abhinav Jain, Cindy Grimm, Stefan Lee. "Learning to Prune Branches in Modern Tree-Fruit Orchards." (ICRA 2025)[\[Paper\]](#)

Deanna Flynn, **Abhinav Jain**, Heather Knight, Christina G. Wilson, Cindy Grimm. "Integrating Stakeholder Perspectives into Robot Pruning Designs". In review [\[Paper\]](#)

Abhinav Jain¹, Dhruv Patel¹, Kalpesh Prajapati, K.P. Upla. "SRTGAN: Triplet Loss based Generative Adversarial Network for Real-World Super-Resolution". [CVIP 2022](#)

Prashant Kumar¹, Sabyasachi Sahoo¹, Vanshil Shah, Vineetha Kondameedi, **Abhinav Jain**, Akshaj Verma, Chiranjib Bhattacharyya, Vinay V. (Sep 2020). "DSLRL: Dynamic to Static LiDAR scan Reconstruction using adversarially trained autoencoder". **AAAI Conference on Artificial Intelligence 2021** (conference submission)[\[web-page\]](#)

LEADERSHIP AND VOLUNTEER EXPERIENCE

President, Robotics Grad Student Association (OSU): Directed the student organization for the 2023-2024 academic year, coordinating events and advocating for graduate students.

Volunteer Editor: Reviewed and refined technical content for the "100-Page LLM Book." [\[Link\]](#)

Invited Speaker: Presented robotics research to a broad audience at the Graduate Research Showcase, a TEDx-style university event. [\[Video\]](#)

Motion Planning Lead, ABU Robocon 2020: Guided the autonomous motion team to achieve **12th** place out of 120 international teams. [\[Video\]](#)

Team Lead, DotSlash Hackathon: Secured Runner-Up by developing "Jyoti," an AI-driven accessibility assistant for visually impaired users. [\[Github\]](#)

Robotics Project: Programmed an industrial Fetch Robot to autonomously complete a scavenger hunt for interactive demonstrations. [\[Video\]](#)

Technical Contributor: Published a predictive modeling analysis on Coronavirus hotspots via the Analytics Vidhya platform. [\[Blog Post\]](#)

TECHNICAL SKILLS

Languages: Python, C, C++, Embedded C, Git

Simulation: NVIDIA Isaac Sim, Isaac Lab, PyBullet, Newton Physics Engine

Frameworks: ROS2, PyTorch, TensorFlow, Keras, Flask, W&B

Libraries: OpenCV, MoveIt!, pandas, NumPy, Matplotlib

Robots worked with: UR5, Fetch, TurtleBot, Franka Panda PR3, XArm7, Amiga FarmNG

¹Equal Contribution